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I confirm that I understand my coursework needs to be submitted online via MST Classroom under the relevant module page before the deadline for my assignment to be accepted and marked. I am fully aware that late submissions will be treated as non-submission and a mark of zero will be awarded.

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1. Introduction

This coursework is a part of the Smart Data Discovery module, which allows us to acquire knowledge and skills on analyzing, preparing, and exploring data through Python programming. The following are some of the major topics discussed in this module:

- Understanding the structure of datasets
- Identifying patterns and relationships between data
- Cleaning up missing or duplicate data
- Making graphs and charts to show results

In our modern world, network security is a biggest challenge. The increasing use of the internet by people and businesses has increased the risk of cyber attacks. Hackers may try to break into computer system to steal information or cause harm. Intrusion Detection Systems (IDS) are used to prevent this. These systems monitor network traffic and find unusual activity that could be an attack. These systems are developed by analyzing and gaining an understanding of large volumes of data.

From this coursework, a real network traffic dataset is used to look at whether network connections are normal or could be an attack. The dataset used is the CICIDS 2017 dataset, which is widely used in cybersecurity research. The work done in this coursework are:

- Data Understanding: The dataset is explored to understand its structure and key features.
- Data Preparation: The data is loaded into Python using pandas. From there, necessary columns are selected, duplicate and missing values are removed, and the labels are renamed to make them easier to understand.
- Data Analysis: Mean, standard deviation, skewness and kurtosis are calculated. The relationships between different variables are also examined using correlation.
- Data Exploration: Bar charts, pie charts, and box plots are drawn, and a hypothesis test is done to check if there is a real difference between normal traffic and attack traffic.

This coursework enhances our understanding of the data and can be applied to prepare the data for further use in the development of machine learning models for network intrusion detection.

1.1. Python and its's Libraries



Figure 1: Python Logo

Python is a popular programming language used in data science and machine learning. There are many libraries available in Python which makes the development process easier and helps in doing different tasks like data analysis and data visualization (Python Software Foundation, 2026). It is open source and has large support, which means there are many resources online that can be used for guidance. In this coursework, Python is used to load, clean, analyze and visualize the network traffic dataset.

Python Libraries

To complete this coursework, several Python libraries are used. They are used for different tasks in data analysis.

Pandas



Figure 2: Pandas Logo

Pandas is a python library that is used for data handling. It provides a structure called a DataFrame, which organizes data in rows and columns, similar to a Microsoft Excel spreadsheet (McKinney, W., 2010). In this coursework, pandas is used to read the dataset, select the columns that are required for the analysis, check and remove any missing values, find and remove duplicate data, and rename the labels to make them easier to understand.

Matplotlib



Figure 3: Matplotlib Logo

Matplotlib is a Python library which is used for creating charts and graphs. It helps to visualize the data in different forms such as bar graphs, pie charts, and box plots (John D. Hunter, 2007). In this coursework, Matplotlib is used to plot bar chart of the frequency of normal and attack traffic, a pie chart of average flow duration and packet length, and a box plot of forward packet length for different traffic types.

Numerical Python (NumPy)



Figure 4: NumPy Logo

NumPy is a Python package used for numerical computing and doing mathematical operations on large arrays (Charles R. Harris *et al.*, 2020). It works well with other libraries like pandas and is commonly used in data science. In this coursework, NumPy is used to help with mathematical calculations, select numeric columns and support functions such as mean and standard deviation for statistical analysis.

Seaborn



Figure 5: Seaborn Logo

Seaborn is a Python visualization library that uses Matplotlib to create statistical plots. The library supports pandas DataFrame, which makes it easy to generate neat and attractive plots since it required minimal code (Michael L. Waskom, 2021). In this coursework, Seaborn is used to create boxplots that help compare the distribution of network traffic features across different categories.

SciPy



Figure 6: SciPy Logo

SciPy is a package in Python that provides tools for scientific and statistical calculations. It is an extended version of NumPy library and offers various functions for conducting statistical operations (Virtanen, P. *et al*, 2020). It is commonly used in data science where more complex calculations are required. In this coursework, SciPy is used to do the hypothesis test to find out if there is a real difference in the mean flow duration between Normal Traffic and Attack traffic.

2. Aim and Objectives

The aim of this coursework is to analyze the CICIDS 2017 network traffic dataset to understand the differences between normal and attack network traffic. This is done by going through the stages of data analysis, starting from loading and cleaning the data to exploring it visually and running statistical tests. The results from this analysis will help us understand network traffic patterns and could support the development of a network intrusion detection system in the future.

The objectives of the coursework are:

- To load the CICIDS 2017 network traffic dataset into python and examine the structure of the data including row count, columns count, data types, and key features.
- To clean the dataset by identifying and eliminating any missing and duplicates to ensure that the information provided is valid and reliable.
- To select only the necessary columns and rename the labels to make them easier to understand.
- To calculate statistics like mean, standard deviation, skewness, and kurtosis to see how the data is spread and spot any unusual values.
- To calculate the correlation between numeric features to identify which variables are correlated.
- To create bar chart, pie chart, and box plot to visually compare normal and attack traffic.
- To run a hypothesis test to check if there is a significant difference between normal and attack traffic based on mean flow duration.
- To use the findings to decide which features are most useful for training machine learning models to detect network attacks in the future.

3. Tools and Technologies

Jupyter Notebook



Figure 7: Jupyter Notebook Logo

Jupyter Notebook is a tool that allows the users to write and execute the Python program step by step in an interactive way. It also has a markdown feature, which helps in adding explanations to the code (Thomas Kluyver *et al.*, 2016). It is popular in data science and research as it allows easy sharing and presentation of work. All the Python programs in this coursework are written and executed in Jupyter Notebook.

Microsoft Word



Figure 8: Microsoft Word Logo

Microsoft Word is a popular writing tool which was developed by Microsoft. It is used to write the technical report for this coursework, because it makes it easy to create well-organized documents with headings, tables, and references (Microsoft Corporation, 2019).

4. Data Understanding

4.1. Overview of Dataset

The data set used in this analysis is based on the CICIDS 2017 (Canadian Institute for Cybersecurity Intrusion Detection System 2017) dataset (Arash Sharafaldin *et al.*, 2018). It contains records of real network traffic, each record is tagged as either BENIGN (normal traffic) or Infiltration (attack traffic).

This dataset is commonly used in cybersecurity research because it helps in building and testing network intrusion detection systems (IDS). In the dataset, every single row stands for one network flow, which is nothing but a connection between two devices that exchange data with each other. These flows are described using various statistical features that are calculated from the packet data.

There are 488,115 rows and 80 columns in dataset, but one of the columns is unnamed and does not contain any useful information, so the actual number of meaningful columns is 79. Each column holds some information about the network traffic, such as the number of packets sent, the duration of the connection, and the amount of data transferred. The dataset is mainly used to train machine learning models that can identify whether network traffic is normal or an attack.

The Label column is the most important column in this dataset as it acts as the target variable. It shows whether a network flow is normal or an attack, which makes it the most important columns when it comes to building an intrusion detection system.

4.2. Dataset Structure

The dataset is loaded into Python using `pd.read_csv()`. The Unnamed:0 column is dropped using `ds.drop()` because does not serve any purpose and only acts as an index column which is not required for any further analysis.

```
# Loading the dataset
# Reading the CSV file into DataFrame
ds = pd.read_csv('ids.csv')

# Dropping unnecessary index column
ds = ds.drop(columns=["Unnamed: 0"])
```

Figure 9: Loading the dataset

Summary statistics of the dataset are calculated by using the function `ds.describe()`. The output shows 488115 records and 78 columns, with key statistics like mean, standard deviation, minimum, and maximum values for each column.

```
# Generating summary statistics
ds.describe()
```

	Destination Port	Flow Duration	Total Fwd Packets	Total Backward Packets	Total Length of Fwd Packets	Total Length of Bwd Packets	Fwd Packet Length Max	Fwd Packet Length Min	Fwd Packet Length Mean	Fwd Packet Length Std	...	act_data_pkt_fwd	mir
count	488115.000000	4.881150e+05	488115.000000	488115.000000	4.881150e+05	4.881150e+05	488115.000000	488115.000000	488115.000000	488115.000000	...	488115.000000	
mean	8216.354384	8.942263e+06	6.206931	6.185579	5.581399e+02	6.222017e+03	140.936187	17.142253	41.737297	40.785650	...	3.436965	
std	17541.650108	2.748091e+07	69.852691	99.359121	2.738195e+04	1.788109e+05	438.404840	31.486744	119.415855	138.185083	...	50.611058	
min	0.000000	-2.000000e+00	1.000000	0.000000	0.000000e+00	0.000000e+00	0.000000	0.000000	0.000000	0.000000	...	0.000000	
25%	53.000000	5.900000e+01	2.000000	1.000000	4.000000e+00	0.000000e+00	2.000000	0.000000	2.000000	0.000000	...	1.000000	
50%	443.000000	3.100000e+02	2.000000	2.000000	4.900000e+01	7.200000e+01	32.000000	2.000000	29.000000	0.000000	...	1.000000	
75%	3221.000000	1.002090e+05	3.000000	2.000000	9.500000e+01	2.460000e+02	51.000000	35.000000	46.000000	0.000000	...	1.000000	
max	65533.000000	1.199999e+08	22673.000000	44553.000000	1.290000e+07	6.360000e+07	23360.000000	2065.000000	4317.097561	4467.113460	...	9130.000000	

8 rows x 78 columns

Figure 10: Generating summary statistics

The shape of the dataset is checked using `ds.shape`. The output shows that the dataset has 488,155 rows and 80 columns, which means there is a large amount of data.

```
# Displaying shape
print(ds.shape)
```

(488115, 79)

Figure 11: Displaying shape

The `ds.info()` function displays detailed information about the dataset. The result indicates that most of the columns contain whole numbers (int64) or decimals numbers (float64). The Label column contains text (object type) that represents normal traffic or attack. The non-null counts shows that most columns do not have any missing values.

```
# Viewing dataset info (columns, dtypes, null values)
ds.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 488115 entries, 0 to 488114
Data columns (total 79 columns):
#   Column                                     Non-Null Count  Dtype
---  -
0   Destination Port                         488115 non-null  int64
1   Flow Duration                           488115 non-null  int64
2   Total Fwd Packets                       488115 non-null  int64
3   Total Backward Packets                  488115 non-null  int64
4   Total Length of Fwd Packets             488115 non-null  int64
5   Total Length of Bwd Packets             488115 non-null  int64
6   Fwd Packet Length Max                   488115 non-null  int64
7   Fwd Packet Length Min                   488115 non-null  int64
8   Fwd Packet Length Mean                  488115 non-null  float64
9   Fwd Packet Length Std                   488115 non-null  float64
10  Bwd Packet Length Max                   488115 non-null  int64
11  Bwd Packet Length Min                   488115 non-null  int64
12  Bwd Packet Length Mean                  488115 non-null  float64
13  Bwd Packet Length Std                   488115 non-null  float64
14  Flow Bytes/s                             488079 non-null  float64
15  Flow Packets/s                           488115 non-null  float64
16  Flow IAT Mean                           488115 non-null  float64
17  Flow IAT Std                             488115 non-null  float64
18  Flow IAT Max                             488115 non-null  int64
19  Flow IAT Min                             488115 non-null  int64
20  Fwd IAT Total                            488115 non-null  int64
21  Fwd IAT Mean                            488115 non-null  float64
22  Fwd IAT Std                             488115 non-null  float64
23  Fwd IAT Max                             488115 non-null  int64
```

Figure 12: Showing info of the dataset (1)

```

24 Fwd IAT Min 488115 non-null int64
25 Bwd IAT Total 488115 non-null int64
26 Bwd IAT Mean 488115 non-null float64
27 Bwd IAT Std 488115 non-null float64
28 Bwd IAT Max 488115 non-null int64
29 Bwd IAT Min 488115 non-null int64
30 Fwd PSH Flags 488115 non-null int64
31 Bwd PSH Flags 488115 non-null int64
32 Fwd URG Flags 488115 non-null int64
33 Bwd URG Flags 488115 non-null int64
34 Fwd Header Length 488115 non-null int64
35 Bwd Header Length 488115 non-null int64
36 Fwd Packets/s 488115 non-null float64
37 Bwd Packets/s 488115 non-null float64
38 Min Packet Length 488115 non-null int64
39 Max Packet Length 488115 non-null int64
40 Packet Length Mean 488115 non-null float64
41 Packet Length Std 488115 non-null float64
42 Packet Length Variance 488115 non-null float64
43 FIN Flag Count 488115 non-null int64
44 SYN Flag Count 488115 non-null int64
45 RST Flag Count 488115 non-null int64
46 PSH Flag Count 488115 non-null int64
47 ACK Flag Count 488115 non-null int64
48 URG Flag Count 488115 non-null int64
49 CNE Flag Count 488115 non-null int64
50 ECE Flag Count 488115 non-null int64
51 Down/Up Ratio 488115 non-null int64
52 Average Packet Size 488115 non-null float64
53 Avg Fwd Segment Size 488115 non-null float64
54 Avg Bwd Segment Size 488115 non-null float64
55 Fwd Header Length.1 488115 non-null int64
56 Fwd Avg Bytes/Bulk 488115 non-null int64
57 Fwd Avg Packets/Bulk 488115 non-null int64
58 Fwd Avg Bulk Rate 488115 non-null int64
59 Bwd Avg Bytes/Bulk 488115 non-null int64
60 Bwd Avg Packets/Bulk 488115 non-null int64
61 Bwd Avg Bulk Rate 488115 non-null int64
62 Subflow Fwd Packets 488115 non-null int64
63 Subflow Fwd Bytes 488115 non-null int64
64 Subflow Bwd Packets 488115 non-null int64
65 Subflow Bwd Bytes 488115 non-null int64
66 Init_Win_bytes_forward 488115 non-null int64
67 Init_Win_bytes_backward 488115 non-null int64
68 act_data_pkt_fwd 488115 non-null int64
69 min_seg_size_forward 488115 non-null int64
70 Active Mean 488115 non-null float64
71 Active Std 488115 non-null float64
72 Active Max 488115 non-null int64
73 Active Min 488115 non-null int64
74 Idle Mean 488115 non-null float64
75 Idle Std 488115 non-null float64
76 Idle Max 488115 non-null int64
77 Idle Min 488115 non-null int64
78 Label 488115 non-null object
dtypes: float64(24), int64(54), object(1)
memory usage: 294.2+ MB

```

Figure 13: Showing info of the dataset (2)

4.3. Description of Columns

The dataset has many columns but only 16 columns are selected for this analysis. Each column represents a different feature of the network traffic flow. Some columns store whole numbers while others store decimal numbers. The last column called Label is a text column that tells us whether the network flow is normal or an attack. It is important to understand what each column means before doing any analysis because it helps to make sense of the results. The table below gives a brief description of each of the 16 selected columns and their data types.

Table 1: Key Columns Description

S.No.	Column Name	Description	Data Type
1	Destination Port	The port number on the destination device, which shows which service the traffic is going to. For example, port 80 is used for normal web browsing and 443 is used for secure web browsing.	Integer
2	Flow Duration	How long the network flow lasted from start to finish, measured in microseconds. A flow that is too long or too short could be a sign of unusual activity.	Integer

3	Total Fwd Packets	The total number of packets sent from the source device to the destination device during the flow. A high number of forward packets may mean that a large amount of data was sent in one direction.	Integer
4	Total Backward Packets	The total number of packets sent back from the destination device to the source device. By comparing it with total forward packets, it can help to understand whether the communication was one sided or two sided.	Integer
5	Total Length of Fwd Packets	The total amount of data in bytes that was sent in the forward direction during the flow. This shows how much data was transferred from the source to the destination.	Integer
6	Total Length of Bwd Packets	The total amount of data in bytes sent in the backward direction during the flow. This shows how much data was sent back from the destination to the source.	Integer
7	Fwd Packet Length Mean	The average size of packets sent from source to destination. This helps to understand the typical size of packets going in the forward direction.	Float
8	Bwd Packet Length Mean	The average size of the packets sent from the destination back to the source. By comparing it with forward packet length mean, it can be used to determine whether the two sides of the connection are sending equal amounts of data.	Float
9	Flow Bytes/s	The number of bytes transferred per second during the flow. A very high value could mean that a large amount of data is	Float

		being sent very quickly, which could be a sign of data theft or a flooding attack.	
10	Flow Packets/s	The number of packets sent per second during the flow. A very high rate can be a sign of a flooding attack or denial of service attack, where the attacker tries to overload the network with too many packets.	Float
11	Packet Length Mean	The average size of all packets in the flow, including both forward and backward packets. This gives a general idea of the typical packet size for the entire connection.	Float
12	Packet Length Std	Shows how much the packet sizes vary from the average. A low value means most packets are a similar size, while a high value means packet sizes are very different from each other. This can help to identify unusual patterns in the network traffic.	Float
13	Average Packet Size	The overall average size of packets in the entire flow. This is similar to Packet Length Mean and gives a general picture of how large the packets are in the connection.	Float
14	Active Mean	The average amount of time in microseconds that the flow was actively sending or receiving data before going quiet. This can help to understand how active the connection was during its lifetime.	Float
15	Idle Mean	The average amount of time in microseconds that the flow was inactive or	Float

		quiet before becoming active again. A high idle time could mean that the connection was paused for a long time between sending data.	
16	Label	The target column of the dataset. It tells us whether the network flow was Normal Traffic or an Attack. This column is used as the dependent variable when training machine learning models for the intrusion detection.	String

5. Data Preparation

The process of cleaning and organizing raw data and then using it to provide an analysis is known as data preparation. Raw data obtained from the real world is messy and cannot be used directly. Data preparation helps to fix these issues so that the results are accurate and reliable (Adriano A. A. Fernandes *et al.*, 2023). Data preparation is therefore an important step in this process.

In this section, the network traffic data is prepared through several steps, which are as follows:

- Loading the data into Python
- Selecting only the useful columns
- Checking and removing missing values
- Checking for duplicate records
- Renaming the label values to make them easier to read

Firstly, all the required Python libraries are imported using the `import` keyword. **Pandas** is used for working with data, **NumPy** is used for numerical calculations, **Matplotlib** is used for creating charts, and **Seaborn** is used for making statistical graphs. Additionally, the `stats` module from **SciPy** is imported to perform statistical tests.

```
# Importing required libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats
```

Figure 14: Importing python libraries

5.1. Load Data and Display Last 50 Records

The first step is to load the data into Python for analysis. This is an important step because if the data is not imported properly, the rest of the steps will be incorrect. It is also a good practice to check some records after loading data to make sure it is correct.

The dataset is loaded from a CSV file called `ids.csv` using the `pd.read_csv()` function and stored in a variable called `df`. The last 50 records are displayed using the `df.tail(50)` to verify the data was loaded correctly. The result shows that the dataset contains 488,115 rows and 80 columns.

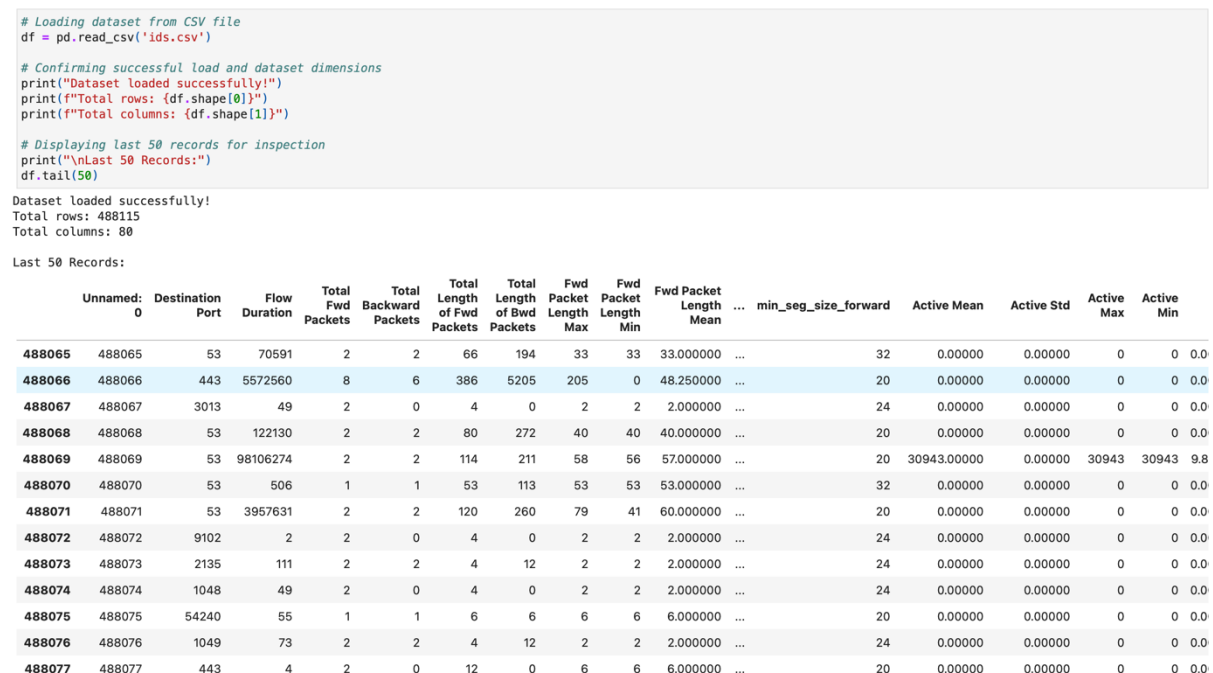


Figure 15: Dataset successfully loaded

5.2. Create a Separate DataFrame with Selected Columns

The dataset contains 80 columns, but only some are required for this analysis. If there are too many columns in the dataset, it can make the analysis more complex and time consuming. If only the necessary columns are used, the analysis becomes more efficient and easier to understand. It also slows to focus on the features that are relevant for monitoring network traffic.

New DataFrame is created as a subset of the original data with only the 16 columns selected as `df_selected`. This simplifies the further analysis of the data.

```
# Defining required feature columns
columns = [
    'Destination Port', 'Flow Duration', 'Total Fwd Packets',
    'Total Backward Packets', 'Total Length of Fwd Packets',
    'Total Length of Bwd Packets', 'Fwd Packet Length Mean',
    'Bwd Packet Length Mean', 'Flow Bytes/s', 'Flow Packets/s',
    'Packet Length Mean', 'Packet Length Std', 'Average Packet Size',
    'Active Mean', 'Idle Mean', 'Label'
]

# Creating a new DataFrame with only the selected columns
df_selected = df[columns]

# Comparing original and subset shapes
print("Original shape:", df.shape)
print("Subset shape:", df_selected.shape)

# Previewing first 5 rows of subset
print("\nFirst 5 rows:")
df_selected.head()
```

Original shape: (488115, 80)
Subset shape: (488115, 16)

First 5 rows:

	Destination Port	Flow Duration	Total Fwd Packets	Total Backward Packets	Total Length of Fwd Packets	Total Length of Bwd Packets	Fwd Packet Length Mean	Bwd Packet Length Mean	Flow Bytes/s	Flow Packets/s	Packet Length Mean	Packet Length Std	Average Packet Size	Active Mean	Idle Mean	Label
0	53	62171	2	2	78	164	39.00	82.000000	3.892490e+03	64.338679	56.200000	23.552070	70.250000	0.00000	0.0	BENIG
1	2710	48	2	0	4	0	2.00	0.000000	8.333333e+04	41666.666670	2.000000	0.000000	3.000000	0.00000	0.0	Infiltrat
2	443	4792909	5	1	135	46	27.00	46.000000	3.776412e+01	1.251849	32.428571	18.866700	37.833333	0.00000	0.0	BENIG
3	80	115596470	75	82	342	118155	4.56	1440.914634	1.025092e+03	1.358173	749.981013	946.124811	754.757962	22092.63636	10000000.0	BENIG
4	2910	14	2	2	4	12	2.00	6.000000	1.142857e+06	285714.285700	3.600000	2.190890	4.500000	0.00000	0.0	BENIG

Figure 16: Creating a separate DataFrame with selected columns

The new DataFrame with 488,115 rows and 16 columns has been successfully created as shown in the output.

5.3. Check and Remove NaN (Missing Values)

Real world data can contain missing values. They can occur due to various reasons including errors in data collection process or issues with the network recording devices. Errors can occur in the analysis and conclusions may be incorrect if missing values are not handled properly. That is why it is important to check the number of missing values and delete or replace them before doing any further analysis.

The `isnull().sum()` method is used to check for missing values in each column. The output shows that only the Flow Bytes/s columns have 36 missing values. There are no missing values in all other columns with a total of 36 missing values in the dataset.

```
# Checking NaN values in each column
print("Missing Values Per Column:\n")
print(df_selected.isnull().sum())

total_missing = df_selected.isnull().sum().sum()
print(f"\nTotal missing values: {total_missing}")
```

Missing Values Per Column:

Destination Port	0
Flow Duration	0
Total Fwd Packets	0
Total Backward Packets	0
Total Length of Fwd Packets	0
Total Length of Bwd Packets	0
Fwd Packet Length Mean	0
Bwd Packet Length Mean	0
Flow Bytes/s	36
Flow Packets/s	0
Packet Length Mean	0
Packet Length Std	0
Average Packet Size	0
Active Mean	0
Idle Mean	0
Label	0
dtype: int64	

Total missing values: 36

Figure 17: Checking NaN (missing values)

Any rows that have missing data are removed using the `dropna()` method to clean the dataset before analysis. After cleaning, the DataFrame now has 488,079 rows and 16 columns, and all columns show 0 missing values, which means that the data is now completely clean and ready for further analysis.

```
# Removing rows with NaN values
df_selected = df_selected.dropna()

# Verifying dataset after removing missing values
print(f"\nDataFrame shape after removing NaN values: {df_selected.shape}\n")
print("Missing values remaining after cleaning:\n")
print(df_selected.isnull().sum())

total_missing = df_selected.isnull().sum().sum()
print(f"\nTotal missing values: {total_missing}")
```

DataFrame shape after removing NaN values: (488079, 16)

Missing values remaining after cleaning:

Destination Port	0
Flow Duration	0
Total Fwd Packets	0
Total Backward Packets	0
Total Length of Fwd Packets	0
Total Length of Bwd Packets	0
Fwd Packet Length Mean	0
Bwd Packet Length Mean	0
Flow Bytes/s	0
Flow Packets/s	0
Packet Length Mean	0
Packet Length Std	0
Average Packet Size	0
Active Mean	0
Idle Mean	0
Label	0
dtype: int64	

Total missing values: 0

Figure 18: Removing NaN (missing values)

5.4. Check for Duplicated Records

Duplicate records are the rows that are repeated in the dataset. Duplicates in the dataset can skew the analysis and influence the results. If there are lots of normal traffic records, then the analysis might think that normal traffic is more frequent than it actually is.

The `duplicated()` method is used to check whether any duplicate records exist in the DataFrame. The output shows that there are 139,389 duplicate records. This is a high value and indicates that a most portion of the dataset contains duplicates. It is important to remove these

duplicates before using the data for training any machine learning model so that the model does not learn from the same data over and over again.

```
# Checking duplicated records and counting them
duplicates_exist = df_selected.duplicated().any()
total_duplicates = df_selected.duplicated().sum()

print(f"Do duplicates exist? {duplicates_exist}")
print(f"Total number of duplicated records: {total_duplicates}")

Do duplicates exist? True
Total number of duplicated records: 139389
```

Figure 19: Checking duplicated records

5.5. Show Total Number of Columns and Their Names

The total number of columns and the names of the columns are printed to know about the data that is being used for the analysis. This is useful to make sure that all the necessary columns are included in the dataset.

The total number of columns is displayed using `len(df_selected.columns)` and all column names are printed using `df_selected.columns`. The result shows that the total number of columns is 16. The columns are clearly listed in order, which provides with an idea of the data available for analysis.

```
# Displaying total number of columns and their names
print("Total number of columns:", len(df_selected.columns))
print("\nColumn Names:\n")
for col in df_selected.columns:
    print(col)

Total number of columns: 16

Column Names:

Destination Port
Flow Duration
Total Fwd Packets
Total Backward Packets
Total Length of Fwd Packets
Total Length of Bwd Packets
Fwd Packet Length Mean
Bwd Packet Length Mean
Flow Bytes/s
Flow Packets/s
Packet Length Mean
Packet Length Std
Average Packet Size
Active Mean
Idle Mean
Label
```

Figure 20: Displaying total number of columns and their names

5.6. Rename Values in the Label Column

The Label is the most significant column in the dataset as it indicates whether a network flow is normal or an attack. But the values of the Label column, BENIGN and Infiltration, are not very clear. It is important to rename them to simpler names to make the analysis and chart interpretation easier.

The values in the Label column are renamed using the `apply()` method with a `lambda` function. BENIGN is renamed to Normal Traffic and Infiltration is renamed to Attack. The output shows that the renaming was completed successfully, with 288,548 Normal Traffic records and 199,531 Attack records, which is the same as the original number of records.

```
# Checking original label values
print("Original Label values:")
print(df_selected['Label'].value_counts())

# Renaming label values
df_selected['Label'] = df_selected['Label'].apply(
    lambda x: 'Normal Traffic' if x == 'BENIGN' else 'Attack'
)

print("\nRenamed Label values:")
print(df_selected['Label'].value_counts())
```

Original Label values:

Label	
BENIGN	288548
Infiltration	199531

Name: count, dtype: int64

Renamed Label values:

Label	
Normal Traffic	288548
Attack	199531

Name: count, dtype: int64

Figure 21: Renaming values in Label column

6. Data Analysis

Data analysis is a process of gathering, cleaning, transforming, and interpreting data to gain useful information for making more effective decisions in different fields. It is highly critical in areas like health care, financial sector, and cybersecurity because it helps to get an insight of what is happening and what action needs to be taken. Data analysis also helps in identifying the distribution of data, relationships between different variables and potential outliers in the dataset (McKinney, W., 2010).

In the field of cybersecurity and network intrusion detection, data analysis is especially important because network traffic contains many different features and it can be hard to know which ones are useful without doing a proper analysis first. It becomes easy to understand the relationship between various elements and the distribution by computing descriptive statistics and correlations between features. This makes it possible to select the most important variables for future modeling and data analysis purposes. In this section, summary statistics of two numerical features are calculated to determine distribution and shape, and correlation between all numeric features is explored.

6.1. Summary Statistics of Two Numeric Variables

Summary statistics is a method of describing a set of data numerically. Here, summary statistics are calculated for two numeric variables: Flow Duration and Packet Length Mean

using `.sum()`, `.mean()`, `.std()`, `.skew()` and `.kurtosis()`. These statistics help in understanding the distribution and shape of these features in the dataset.

```
# Summary statistics of two numeric column (Flow Duration and Packet Length Mean)
# Sum of selected columns
print("\nSum of each column:")
print(df_selected[['Flow Duration', 'Packet Length Mean']].sum())

# Mean of selected columns
print("\nMean of Flow Duration & Packet Length Mean:")
print(df_selected[['Flow Duration', 'Packet Length Mean']].mean())

# Standard deviation of selected columns
print("\nStandard Deviation of Flow Duration & Packet Length Mean:")
print(df_selected[['Flow Duration', 'Packet Length Mean']].std())

# Skewness (asymmetry of distribution)
print("\nSkewness of Flow Duration & Packet Length Mean:")
print(df_selected[['Flow Duration', 'Packet Length Mean']].skew())

# Kurtosis (peakedness of distribution)
print("\nKurtosis of Flow Duration & Packet Length Mean:")
print(df_selected[['Flow Duration', 'Packet Length Mean']].kurtosis())
```

Sum of each column:

Flow Duration	4.364853e+12
Packet Length Mean	4.175763e+07

dtype: float64

Mean of Flow Duration & Packet Length Mean:

Flow Duration	8.942923e+06
Packet Length Mean	8.555506e+01

dtype: float64

Standard Deviation of Flow Duration & Packet Length Mean:

Flow Duration	2.748182e+07
Packet Length Mean	1.738725e+02

dtype: float64

Skewness of Flow Duration & Packet Length Mean:

Flow Duration	3.250874
Packet Length Mean	4.935766

dtype: float64

Kurtosis of Flow Duration & Packet Length Mean:

Flow Duration	9.246791
Packet Length Mean	34.610512

dtype: float64

Figure 22: Summary Statistics for two numeric variables

For Flow Duration, the total sum is 4.364853e+12 and the mean is 8.942923e+06. For the skewness, the obtained result is 3.250874. Being positive, it implies that the variable's distribution is skewed to the right and that its values are concentrated on the left side of the mean. For kurtosis, the obtained value is 9.246791. A large number shows that some extremely high values exist in the data.

For Packet Length Mean, the total sum of all values is 4.175763e+07 and the mean is 8.555506e+01. Skewness is also positive and equals 4.935766. It means that, in the data, there are small packets but there are also very big packets which makes the skewness positive. For the variable, the obtained kurtosis is equal to 34.610512. The kurtosis is very large and indicates that there are extreme values in the data.

6.2. Correlation of All Numeric Variables

Correlation analysis is used to determine the relationship between two variables. In case there is a strong correlation between the two variables, an increase in one variable results in an increase in the other variable as well. In case of a negative correlation, an increase in one variable leads to decrease in the other variable. Correlation values range from -1 to 1, where 1

means a perfect positive relationship, -1 means a perfect negative relationship, and 0 means no relationship at all.

In this analysis, the correlation between all numeric variables is calculated using the `corr()` method with `numeric_only=True` to make sure that the correlations are only calculated for number based columns. The top 5 most correlated feature pairs are then identified by removing self-correlations and duplicate pairs. The remaining pairs are then sorted by their correlation value to find the strongest relationships.

```
# Calculating correlation matrix (numeric columns only)
correlation_matrix = df_selected.corr(numeric_only=True)

# Finding top 5 correlated feature pairs (excluding self-correlation)
corr_pairs = (
    correlation_matrix
    .abs()
    .unstack()
    .reset_index()
)
corr_pairs.columns = ['Feature 1', 'Feature 2', 'Correlation']

# Removing self-correlations and duplicate pairs
corr_pairs = corr_pairs[
    corr_pairs['Feature 1'] < corr_pairs['Feature 2']
].sort_values('Correlation', ascending=False)

print("Top 5 Correlated Feature Pairs:\n")
print(corr_pairs.head(5).to_string(index=False))
```

Top 5 Correlated Feature Pairs:

Feature 1	Feature 2	Correlation
Average Packet Size	Packet Length Mean	0.995068
Total Backward Packets	Total Length of Bwd Packets	0.961480
Total Backward Packets	Total Fwd Packets	0.949848
Total Fwd Packets	Total Length of Bwd Packets	0.920007
Packet Length Mean	Packet Length Std	0.887792

Figure 23: Top 5 correlated feature pairs

The results show that Average Packet Size and Packet Length Mean have the highest correlation of 0.995068. Total Backward Packets and Total Length of Bwd Packets have a correlation of 0.961480, and Total Backward Packets and Total Fwd Packets have a correlation of 0.949848. Total Fwd Packets and Total Length of Bwd Packets have a correlation of 0.920007, and Packet Length Mean and Packet Length Std have a correlation of 0.887792. These high correlations suggest that some features have very similar information and can be removed during feature selection in machine learning.

7. Data Exploration

Data exploration is the step where the dataset is visualized and analyzed to understand it better. Data exploration does not only aid in preparing the dataset but also helps in making decisions, detecting financial frauds, decreasing risks in healthcare, data quality, and maintaining flexibility in the organization to function well in dynamic environments (Taboada, G. L. *et al.*,

2020). In this section, visualization tools such as bar charts, pie charts, and boxplots are used to explore the network traffic dataset.

7.1. Bar Chart: Frequency of Each Category in the Label Column

Bar chart is an effective visual tool that uses rectangular bars to show the frequency or count of each category in a dataset. They are very useful for comparing the categories due to its clear structure (Atlassian, 2026). In this analysis, a bar chart is used to show the amount of Normal Traffic and Attack traffic in the dataset.

A bar chart is plotted using `.value_counts().plot()` to show how many records belong to each category in the Label column. This helps to see if the dataset is balanced or not between Normal Traffic and Attack traffic.

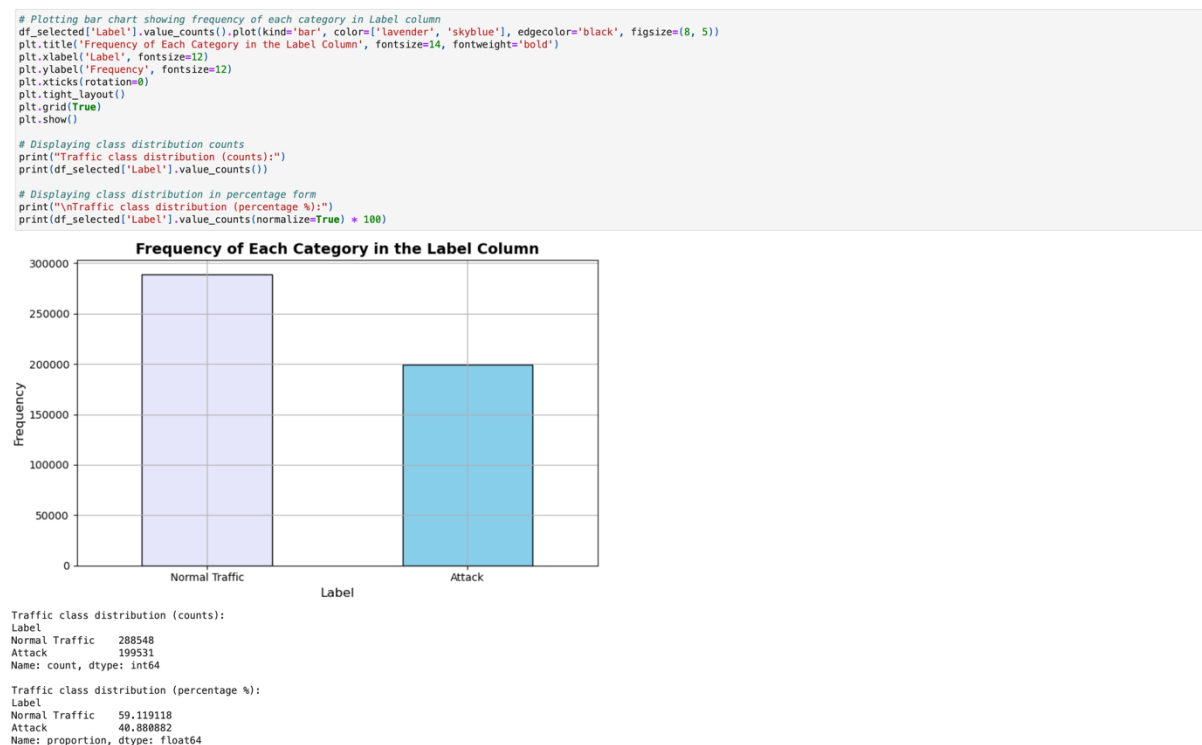


Figure 24: Bar chart showing frequency of each category in the Label column

The bar chart shows that most of the records in dataset are Normal Traffic, with 288548 records, which makes up around 59.1% of the data. Attack traffic is lower, with about 199531 records (around 40.9%). The difference between the two is not huge, but normal traffic is still clearly higher than attack traffic. Such type of difference is quite typical in real world network data. As most of the internet traffic on any given day is likely to be legitimate. The problem is that when a machine learning model is trained on this data, it may lean towards predicting Normal

Traffic more often and miss actual attacks, even if it looks like it is performing well. To fix this, methods like SMOTE, which creates extra attack samples, or undersampling, which cuts down the number of Normal Traffic records, could be applied before training a model. So the dataset is not perfectly balanced because there are more Normal Traffic records than Attack traffic records.

7.2. Pie Chart: Average Flow Duration and Packet Length Mean by Label

Pie chart is a circular chart, which is divided into different slices. They are very useful for comparing proportions between categories (Jaspersoft, 2026). In this analysis, two pie charts are used to compare Flow Duration and average Packet Length Mean between Normal Traffic and Attack traffic.

The average Flow Duration and average Packet Length Mean for each label class are calculated using `groupby()` and `.mean()`. Two pie charts are then plotted next to each other using `average_values.plot()` to compare these averages between Normal Traffic and Attack traffic.

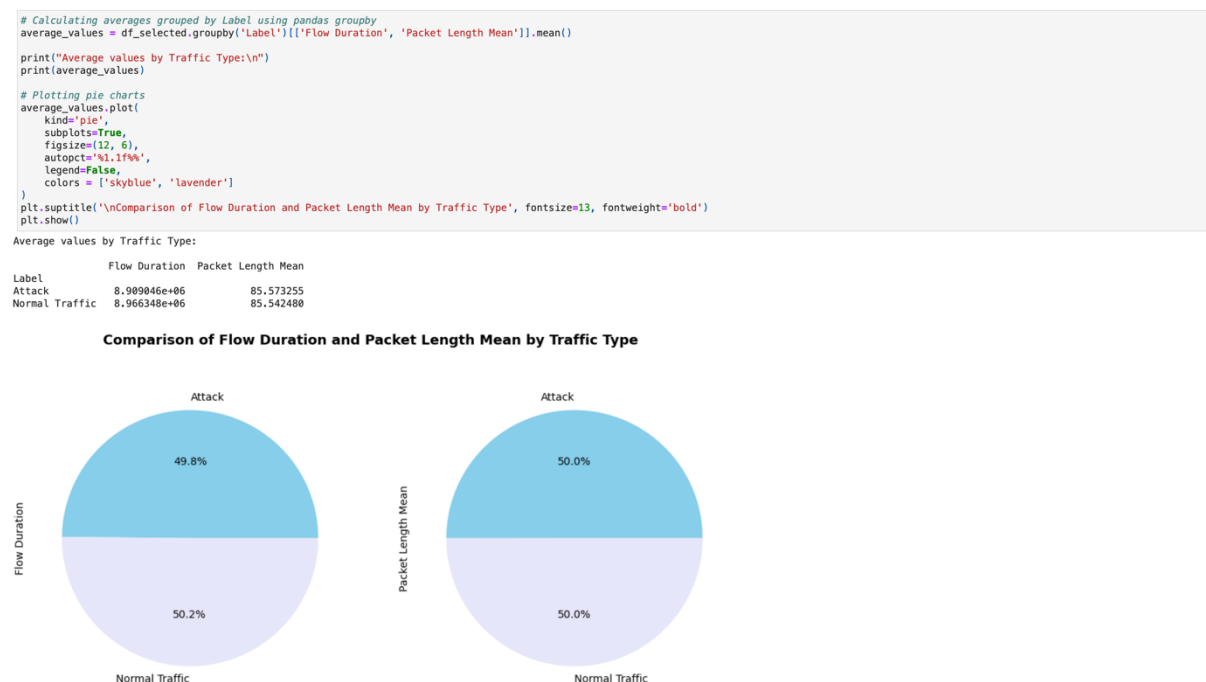


Figure 25: Pie chart showing average flow duration and packet length mean

From the above pie charts, Normal Traffic seems to be leading by having the highest average Flow Duration which stands at about $8.966348e+06$ and occupies 50.2% of the first pie. This is not surprising considering that normal activities such as lengthy web browsing, background

running of applications, and consistent file downloading make the connections remain open for longer. The proportion of Attack Traffic is 49.8%, which is slightly lower than that of Normal Traffic. In second pie chart, the Attack Traffic has a mean Packet Length of 88.573255 bytes, whereas the Normal Traffic has 85.542480. Even though the pie looks almost evenly split, the numbers printed above confirm that attack flows do carry slightly larger packets on average. The difference is only 0.03 bytes which is very small. But Flow Duration and Packet Length do show some differences between the two types of traffic. These differences might be useful when building a model to detect intrusions like an intrusion detection model, for Flow Duration and Packet Length.

7.3. Box Plot: Fwd Packet Length Mean Grouped by Label

Box plot is a type of chart that shows the distribution and range of data values. It also shows outliers, which are values that are very different from the rest. Box plots are very useful when comparing the spread of values between two or more categories (GeeksforGeeks, 2025).

A boxplot is plotted to show the distribution of Fwd Packet Length Mean for each label category using `sns.boxplot()`. This helps in the comparison of the dispersion of forward packet length between Normal Traffic and Attack traffic.

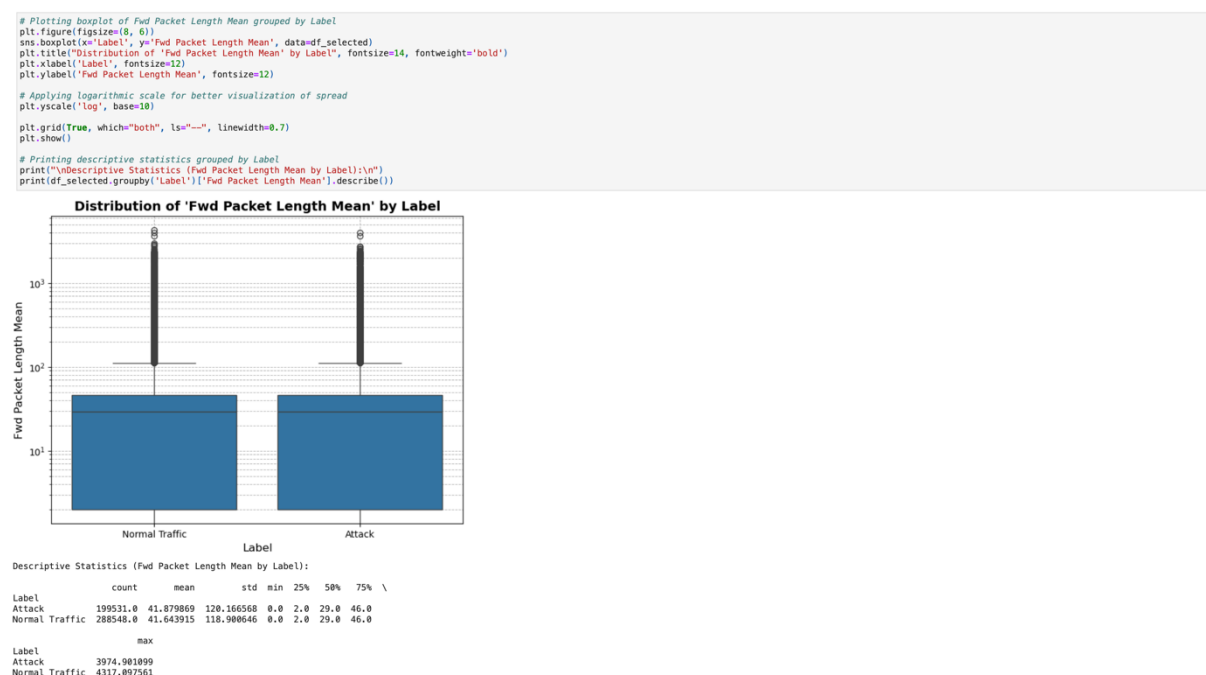


Figure 26: Box plot showing Fwd packet length mean grouped by label

Looking at the boxplot, both Normal Traffic and Attack traffic look almost the same. Both boxes sit close to zero, which means that the forward packet lengths are small for most flows, regardless of whether the traffic is normal or attack. The Attack traffic has an average of 41.88 bytes and Normal Traffic has an average of 41.64 bytes. The difference is very small and does not really matter in the real world. There are some very high outliers in both categories, with Normal Traffic reaching up to 4317.10 bytes and Attack traffic reaching up to 3974.09 bytes. Normal Traffic has more of these high values, which could be because of things like video streaming or large file downloads. Since both traffic types look so alike, Fwd Packet Length Mean is probably not a good feature on its own for separating Normal Traffic from Attack traffic. But when it is combined with other features inside a machine learning model, the outlier behaviour within this variable could be useful for identifying suspicious or potentially harmful network activity.

8. Statistical Testing

Statistical testing can be defined as a process or approach where decisions about a process are made quantitatively based on whether there exists enough evidence to reject or support the expressed hypothesis called the null hypothesis (H_0). The process tests whether there exists any significant difference between two data groups (Virtanen, P. *et al.*, 2020). In this section, the hypothesis test is carried out to test the difference in the mean Flow Duration for Normal Traffic and Attack traffic.

8.1. Hypothesis Test: Mean Flow Duration Between Classes

A hypothesis test is done to see if there exists statistically significant difference in the mean Flow Duration for Normal Traffic and Attack traffic using `stats.ttest_ind()`.

The two hypotheses for this test are:

- Null Hypothesis (H_0): There is no difference in the mean Flow Duration between Normal Traffic and Attack Traffic.
- Alternative Hypothesis (H_1): There is a difference in the mean Flow Duration between Normal Traffic and Attack Traffic.

```

# Hypothesis test: comparing Flow Duration between traffic classes
print("Hypothesis Test: Independent Samples T-Test")
print("====")
Null Hypothesis (H0) : There is NO significant difference in mean Flow Duration between Normal Traffic and Attack traffic.
Alternative Hypothesis (H1): There IS a significant difference in mean Flow Duration between Normal Traffic and Attack traffic.
Significance Level (alpha) : 0.05
Test Chosen : Welch's Independent Samples T-Test
Reason : The two groups are independent from each other and the sample size is large enough for a t-test.
=====

# Separating data by traffic class
normal_flow = df_selected[df_selected['Label'] == 'Normal Traffic']['Flow Duration']
attack_flow = df_selected[df_selected['Label'] == 'Attack']['Flow Duration']

# Performing Welch's t-test
t_stat, p_value = stats.ttest_ind(normal_flow, attack_flow, equal_var=False)

print("T-statistic:", round(t_stat, 4))
print("P-value:", round(p_value, 6))

alpha = 0.05

# Interpreting results
print("\nInterpretation:")
if p_value <= alpha:
    print("Result: REJECT the Null Hypothesis (H0).")
    print("Conclusion: There IS a statistically significant difference in mean Flow Duration between Normal Traffic and Attack classes.")
else:
    print("Result: FAIL TO REJECT the Null Hypothesis (H0).")
    print("Conclusion: There is NO statistically significant difference in mean Flow Duration between Normal Traffic and Attack classes.")

Hypothesis Test: Independent Samples T-Test

Null Hypothesis (H0) : There is NO significant difference in mean Flow Duration between Normal Traffic and Attack traffic.
Alternative Hypothesis (H1): There IS a significant difference in mean Flow Duration between Normal Traffic and Attack traffic.
Significance Level (alpha) : 0.05
Test Chosen : Welch's Independent Samples T-Test
Reason : The two groups are independent from each other and the sample size is large enough for a t-test.

T-statistic: 0.7166
P-value: 0.473601

Interpretation:
Result: FAIL TO REJECT the Null Hypothesis (H0).
Conclusion: There is NO statistically significant difference in mean Flow Duration between Normal Traffic and Attack classes.

```

Figure 27: Hypothesis test involving mean flow duration between classes

Welch's independent samples t-test was chosen because the two groups, Normal Traffic and Attack, are completely independent of each other, and also there is sufficient data available in the dataset for conducting the test. As a result of the higher p-value than 0.05, the null hypothesis cannot be rejected, which means that there is no statistically significant difference in the mean Flow Duration for the two types of traffic. This implies that the Flow Duration for normal and malicious flows behaves identically. Thus, Flow Duration may not be a strong feature on its own in classifying these two types of flows in an intrusion detection system and must be considered along with other features.

9. Conclusion

9.1. Summary of Work Done

In this report, all the required stages of analysis of the network traffic dataset were performed. During data understanding phase, explorations were conducted on the dataset to gain some insights into its contents. During data preparation phase, the dataset was loaded in Python, the necessary columns were selected, and missing and duplicates were dropped, along with renaming of the labels to Normal Traffic and Attack traffic. During data analysis phase, statistical measures were calculated on Flow Duration and Packet Length Mean columns. Finally, during the data exploration stage, graphs were plotted comparing the Normal Traffic and Attack traffic, and a hypothesis test was performed to find out whether the difference in the average Flow Duration of two types of traffic is real or accidental.

9.2. Key Findings

The analysis showed many interesting results. The bar chart showed that the dataset is slightly imbalanced with Normal Traffic making up about 59% and Attack traffic making up about 41% which needs to be considered before training any machine learning model. The pie chart and box plot showed that the average Flow Duration, Packet Length Mean, and Fwd Packet Length Mean are very similar between the two traffic types. The correlation analysis showed that some features are very closely related to each other and could be removed to simplify future models. The hypothesis test gave a p-value of 0.473601, which is greater than 0.05. This means that there is no difference in the mean Flow Duration between Normal Traffic and Attack traffic.

9.3. What was Learned

This coursework helped to understand how to work with real world data using Python. It showed how important it is to clean and prepare the data properly before doing any analysis because messy data can lead to wrong results. It also demonstrated the usefulness of charts in identifying patterns in the data, and how hypothesis testing is helpful in determining whether differences in the data are inherent or merely occurred by chance. Overall, this coursework gave a good understanding of the key steps in data analysis and smart data discovery, and the work done here prepares the data well for the next stage where machine learning models can be built to detect network attacks.

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